Task 2: Paninian to Universal Dependency Converter

The assignment requires you to write a Python script that converts Paninian Dependency trees into Universal Dependency (UD) trees. Your script must accept two command-line arguments (input and output JSON files) and preserve the provided code skeleton.

Here is the fully implemented converter.py script. The dictionary maps core arguments (like *k1* and *k2*), adjuncts, and modifiers, fulfilling the evaluation criteria:

Python

```
import sys
import json

def map_relation(paninian_tag):
    """
    Maps a single Paninian dependency tag to a UD tag.
    Returns 'dep' as a fallback for unknown/unmapped tags.
    """
    # Standard Paninian to UD mapping logic
    mapping_dict = {
        'k1': 'nsubj',        # Karta (Agent/Subject)
        'k2': 'obj',          # Karma (Patient/Object)
        'k3': 'obl:inst',     # Karana (Instrument)
        'k4': 'iobj',         # Sampradana (Beneficiary/Recipient)
        'k5': 'obl',          # Apadana (Source)
        'k6': 'nmod:poss',    # Sambandha (Genitive/Possessor)
        'k7': 'obl',          # Adhikarana (General Location/Time)
        'k7p': 'obl:loc',     # Adhikarana (Place)
        'k7t': 'obl:tmod',    # Adhikarana (Time)
        'r6': 'nmod',         # Relational modifier
        'ccof': 'conj',       # Co-ordination
        'vmod': 'advmod',     # Verb modifier
        'nmod': 'nmod',       # Noun modifier
        'jjmod': 'amod',      # Adjective modifier
        'pof': 'compound:lvc',# Part of complex predicate
        'root': 'root',       # Root node
        'main': 'root'        # Main verb
    }

    # Clean the input tag to handle potential casing/spacing edge cases
    clean_tag = str(paninian_tag).strip().lower()

    # Return the mapped UD tag, falling back to 'dep' as instructed
    return mapping_dict.get(clean_tag, 'dep')

def process_file(input_path, output_path):
```

```
    """
    Reads the input JSON, applies the mapping, and writes to output JSON.
    """
    with open(input_path, 'r', encoding='utf-8') as f:
        tree_data = json.load(f)

    for node in tree_data:
        # Extract the Paninian tag and find the UD equivalent
        p_tag = node.get('paninian_relation', '')
        node['ud_relation'] = map_relation(p_tag)

    with open(output_path, 'w', encoding='utf-8') as f:
        json.dump(tree_data, f, indent=4)

if __name__ == "__main__":
    # Do NOT modify this execution block
    if len(sys.argv) != 3:
        print("Usage: python converter.py <input.json> <output.json>")
        sys.exit(1)

    input_file = sys.argv[1]
    output_file = sys.argv[2]
    process_file(input_file, output_file)
```

Task 1: Syntactic Trees Example

The assignment requires you to analyze ten specific sentences across three frameworks. Here is how you would map out **Sentence 5 (S5): "Lazy men write assignments lazily."**.

1. X-bar Schema (Bracketed Structure) The X-bar schema relies on hierarchical internal structures featuring specifier, head, and complement positions.

- `[IP [NP [AP Lazy] [N' men]] [I' [I Ø] [VP [V' [V' write] [NP assignments]] [AdvP lazily]]]]`

2. Universal Dependencies (UD) UD relates words to a head via a typed arc.

- **write** -> root (Head: 0)
- **men** -> nsubj (Head: write)
- **Lazy** -> amod (Head: men)
- **assignments** -> obj (Head: write)
- **lazily** -> advmod (Head: write)

3. Paninian Dependency (PD) PD annotates relations using *kārakas* (semantic participant roles).

- **write** -> root / main (Head: 0)
- **men** -> k1 (Karta / Doer) (Head: write)
- **Lazy** -> jjmod (Adjectival modifier) (Head: men)
- **assignments** -> k2 (Karma / Patient) (Head: write)
- **lazily** -> vmod (Verb modifier) (Head: write)

Here is the complete structural breakdown for all 10 sentences across the X-bar Schema, Universal Dependencies (UD), and Paninian Dependencies (PD) frameworks.

The dependency analyses map the words to their appropriate grammatical relations and identify their parent nodes (Heads).

S1: The red apples were washed thoroughly, but the green apples were washed badly.

1. X-bar Schema

[CP [IP [NP The red [N' apples]] [I' were [VP washed thoroughly]]] [Conj but] [IP [NP the green [N' apples]] [I' were [VP washed badly]]]]

2. Universal Dependencies (UD)

- **washed (1)** -> root (Head: 0)
- apples (1) -> nsubj:pass (Head: washed 1)
- The, red -> det, amod (Head: apples 1)
- were (1) -> aux:pass (Head: washed 1)
- thoroughly -> advmod (Head: washed 1)
- but -> cc (Head: washed 2)
- **washed (2)** -> conj (Head: washed 1)
- apples (2) -> nsubj:pass (Head: washed 2)
- the, green -> det, amod (Head: apples 2)
- were (2) -> aux:pass (Head: washed 2)
- badly -> advmod (Head: washed 2)

3. Paninian Dependency (PD)

- **washed (1)** -> root (Head: 0)
- apples (1) -> k2 (Karma/Patient) (Head: washed 1)
- thoroughly -> vmod (Verb modifier) (Head: washed 1)
- but -> ccof (Conjunction) (Head: washed 1)
- **washed (2)** -> ccof (Head: but)
- apples (2) -> k2 (Karma/Patient) (Head: washed 2)
- badly -> vmod (Verb modifier) (Head: washed 2)

S2: Priya told the police that she was the perpetrator.

1. X-bar Schema

[IP [NP Priya] [VP told [NP the police] [CP that [IP [NP she] [VP was [NP the perpetrator]]]]]]

2. Universal Dependencies (UD)

- **told** -> root (Head: 0)
- Priya -> nsubj (Head: told)
- police -> iobj (Head: told)
- perpetrator -> ccomp (Head: told)
- that -> mark (Head: perpetrator)
- she -> nsubj (Head: perpetrator)
- was -> cop (Head: perpetrator)

3. Paninian Dependency (PD)

- **told** -> root (Head: 0)
- Priya -> k1 (Karta) (Head: told)
- police -> k4 (Sampradana/Recipient) (Head: told)
- perpetrator -> k2 (Karma clause) (Head: told)
- she -> k1 (Head: perpetrator)
- was -> aux (Head: perpetrator)

S3: Ram’s tables are wooden.

1. X-bar Schema

[IP [NP [NP Ram's] tables] [I' are [AP wooden]]]

2. Universal Dependencies (UD)

- **wooden** -> root (Head: 0) *(Note: UD treats the predicate as the root in copular sentences)*
- tables -> nsubj (Head: wooden)
- Ram's -> nmod:poss (Head: tables)
- are -> cop (Head: wooden)

3. Paninian Dependency (PD)

- **wooden** -> root (Head: 0)
- tables -> k1 (Karta) (Head: wooden)
- Ram's -> r6 (Sambandha/Genitive) (Head: tables)
- are -> aux (Head: wooden)

S4: I am tired because I did not sleep well yesterday.

1. X-bar Schema

[IP [IP [NP I] [I' am [AP tired]]] [CP because [IP [NP I] [I' did not [VP sleep well yesterday]]]]]

2. Universal Dependencies (UD)

- **tired** -> root (Head: 0)
- I (1) -> nsubj (Head: tired)
- am -> cop (Head: tired)
- sleep -> advcl (Head: tired)
- because -> mark (Head: sleep)
- I (2) -> nsubj (Head: sleep)
- did -> aux (Head: sleep)
- not -> advmod (Head: sleep)
- well -> advmod (Head: sleep)
- yesterday -> obl:tmod (Head: sleep)

3. Paninian Dependency (PD)

- **tired** -> root (Head: 0)
- I (1) -> k1 (Karta) (Head: tired)
- sleep -> rs (Relation marker/Adverbial) (Head: tired)
- I (2) -> k1 (Karta) (Head: sleep)
- well -> vmod (Head: sleep)
- yesterday -> k7t (Adhikarana/Time) (Head: sleep)

S5: Lazy men write assignments lazily.

1. X-bar Schema

[IP [NP [AP Lazy] [N' men]] [I' [I Ø] [VP [V' [V' write] [NP assignments]] [AdvP lazily]]]]

2. Universal Dependencies (UD)

- **write** -> root (Head: 0)
- men -> nsubj (Head: write)
- Lazy -> amod (Head: men)
- assignments -> obj (Head: write)
- lazily -> advmod (Head: write)

3. Paninian Dependency (PD)

- **write** -> root (Head: 0)
- men -> k1 (Karta) (Head: write)

- Lazy -> jjmod (Head: men)
- assignments -> k2 (Karma) (Head: write)
- lazily -> vmod (Head: write)

S6: Bananas and guavas will be cut by Rahul for the fruit salad.

1. X-bar Schema

[IP [NP Bananas and guavas] [I' will be [VP cut [PP by Rahul] [PP for the fruit salad]]]]

2. Universal Dependencies (UD)

- **cut** -> root (Head: 0)
- Bananas -> nsubj:pass (Head: cut)
- and -> cc (Head: guavas)
- guavas -> conj (Head: Bananas)
- will, be -> aux, aux:pass (Head: cut)
- Rahul -> obl:agent (Head: cut)
- by -> case (Head: Rahul)
- salad -> obl (Head: cut)
- for -> case (Head: salad)

3. Paninian Dependency (PD)

- **cut** -> root (Head: 0)
- Bananas, guavas -> k2 (Karma) (Head: cut) *(coordinated)*
- Rahul -> k1 (Karta/Agent) (Head: cut)
- salad -> rt (Purpose) (Head: cut)

S7: He thought he could escape the prison.

1. X-bar Schema

[IP [NP He] [I' thought [CP [IP [NP he] [I' could [VP escape [NP the prison]]]]]]]

2. Universal Dependencies (UD)

- **thought** -> root (Head: 0)
- He (1) -> nsubj (Head: thought)
- escape -> ccomp (Head: thought)
- he (2) -> nsubj (Head: escape)
- could -> aux (Head: escape)
- prison -> obj (Head: escape)

3. Paninian Dependency (PD)

- **thought** -> root (Head: 0)
- He (1) -> k1 (Karta) (Head: thought)
- escape -> k2 (Karma clause) (Head: thought)
- he (2) -> k1 (Head: escape)
- prison -> k5 (Apadana/Source of escaping) (Head: escape)

S8: Men are from Mars and women are from Venus.

1. X-bar Schema

[CP [IP [NP Men] [I' are [PP from Mars]]] [Conj and] [IP [NP women] [I' are [PP from Venus]]]]

2. Universal Dependencies (UD)

- **Mars** -> root (Head: 0) *(Predicate is root)*
- Men -> nsubj (Head: Mars)
- from -> case (Head: Mars)
- and -> cc (Head: Venus)
- **Venus** -> conj (Head: Mars)
- women -> nsubj (Head: Venus)
- from -> case (Head: Venus)

3. Paninian Dependency (PD)

- **Mars** -> root (Head: 0)
- Men -> k1 (Karta) (Head: Mars)
- and -> ccof (Head: Mars)
- **Venus** -> ccof (Head: and)
- women -> k1 (Head: Venus)

S9: The teacher made the students read the chapter.

1. X-bar Schema

[IP [NP The teacher] [I' made [VP [NP the students] [V' read [NP the chapter]]]]]

2. Universal Dependencies (UD)

- **made** -> root (Head: 0)
- teacher -> nsubj (Head: made)
- students -> obj (Head: made)
- read -> xcomp (Head: made)
- chapter -> obj (Head: read)

3. Paninian Dependency (PD) *(Causative Frame)*

- **read** -> root (Head: 0)
- teacher -> pk1 (Prayojaka Karta / Causer) (Head: read)
- students -> k1 (Prayojya Karta / Causee) (Head: read)
- chapter -> k2 (Karma) (Head: read)
- made -> pof (Causative marker) (Head: read)

S10: The cake was baked by Meera and decorated by her sister.

1. X-bar Schema

[IP [NP The cake] [I' was [VP [VP baked [PP by Meera]] [Conj and] [VP decorated [PP by her sister]]]]]

2. Universal Dependencies (UD)

- **baked** -> root (Head: 0)
- cake -> nsubj:pass (Head: baked)
- Meera -> obl:agent (Head: baked)
- and -> cc (Head: decorated)
- **decorated** -> conj (Head: baked)
- sister -> obl:agent (Head: decorated)

3. Paninian Dependency (PD)

- **baked** -> root (Head: 0)
- cake -> k2 (Karma) (Head: baked)
- Meera -> k1 (Karta/Agent) (Head: baked)
- and -> ccof (Head: baked)
- **decorated** -> ccof (Head: and)
- sister -> k1 (Karta) (Head: decorated)
- her -> r6 (Sambandha) (Head: sister)